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AGNI: A radiative-convective model for lava planet atmospheres.

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Summary

It is important that we are able to accurately model the atmospheres of (exo)planets. This is because their atmospheres play a crucial role in setting the environment and conditions on the planet, including how the planet evolves over astronomical timescales. Additionally, it is primarily by observation of their atmospheres that we are able to characterise exoplanets. There is demand for accurate atmosphere models in the context of lava worlds: planets with permanent or fleeting magma oceans.

AGNI[1] is a Julia program designed to solve for the temperature and radiation environment within the atmospheres of rocky (exo)planets. It leverages a well established FORTRAN code to calculate radiative fluxes from a given atmospheric temperature structure and composition, which – alongside representations of convection and other processes – enables an energy-conserving numerical solution for the atmospheric conditions. In contrast to most other numerical atmosphere codes, AGNI uses an Newton-Raphson optimisation method to obtain this solution which enables improved performance and scalability. AGNI was specifically developed for use alongside planetary interior models within a coupled framework, although it can also be easily applied to scientific problems as a standalone code.

AGNI can be interacted with as a library; it is used in this way within the Jupyter notebook tutorials in the repository. It can also be used as an executable program, where it reads TOML configuration files from the disk and outputs figures and NetCDF data to a specified directory. The documentation can be read online here.

Statement of need

It is thought that all rocky planets go through a "magma ocean" stage, where their mantles are fully molten (Elkins-Tanton & Seager, 2008; Lichtenberg & Miguel, 2024; Schaefer et al., 2016). For some planets this may be their permanent state, while for others it is fleeting. This stage allows for rapid exchange of energy and material between their atmospheres and interiors. Since this phase is likely common to many planets – including Earth and Venus – it is important that we understand the physical processes involved, and how these processes interact with each other (Maurice et al., 2024; Schaefer & Fegley, 2017). Accurate atmosphere models can allow us to connect the theory of these young planets to telescope observations, since it is primarily through their atmospheric properties that were are able to characterise them (Perryman, 2018; Piette et al., 2023). Modelling these young planets involves facing several poorly contrained quantites which govern their atmospheric composition (Guimond et al., 2023; Sossi et al., 2020). Recently, combined observations and modelling of exoplanet TRAPPIST-1 b enabled inference of its past climactic history (Zieba et al., 2023). Hu et

[1] AGNI can be found on GitHub here.

al. (2024) were able to characterise the atmosphere of 55 Cancri e by using an advanced proprietary atmosphere model.

Several theoretical studies have modelled the atmospheres and evolution of these young planets, but all have made several simplifying assumptions. Lichtenberg et al. (2021) coupled a simple atmosphere model to real-gas radiative transfer and interior evolution codes to simulate magma ocean evolution, but did not account for the possibility of convective stability in their atmospheres. Krissansen-Totton et al. (2021) made similar assumptions. Selsis et al. (2023) used a pure-steam radiative-convective model to investigate convective stability, finding that it is likely to occur, but did not extend their work to coupled scenarios which explore the secular evolution of these planets or gas mixtures. It is unlikely that these atmospheres are exclusively composed of steam (Lichtenberg et al., 2021). Zilinskas et al. (2023) used a multi-gas model to simulate the atmospheres of Sub-Neptune and Super-Earth exoplanets an equilibrium, finding that the opacity of various gaseous species (notably SiO) plays a key role in determining the structure and observable properties of exoplanet atmospheres. The demand for realistic atmosphere modelling in the context of secular magma ocean evolution is apparent.

Ensuring sufficient spectral resolution is important in modelling the blanketing effect of these atmospheres, as resolving the opacity (and transparency) of their many gases is known to be key in setting the rate at which these planets can cool by radiation to space (Boukrouche et al., 2021; Pierrehumbert, 2010). It is also important that we are able to run grids of models which explore the range of possible (and as-yet poorly constrained) conditions that these planets could exhibit, which demands efficient modelling given finite computational resources. Magma ocean crystallisation could take up to several Gyr due to continuous tidal forcing and atmospheric blanketing (Driscoll & Barnes, 2015; Walterova & Behounkova, 2020). The efficiency afforded by AGNI enables Gyr simulations with coupled interior-atmosphere magma ocean models.

Comparison with other codes

AGNI is a new radiative-convective atmosphere model developed with the view of being coupled into the PROTEUS framework[2] alongside other physical models. In solving the aforementioned problems, it is able to:

- be coupled to an interior model with an appropriate surface boundary condition,
- account for atmospheres of mixed gaseous composition with realistic opacities,
- solve for an atmospheric temperature structure which conserves energy and allows for convective stability,
- operate with sufficient speed that it may be participate in a wide parameter space,
- constrain convective mass diffusivity using mixing length theory.

This is possible due to the method by which AGNI numerically obtains a solution for atmospheric temperature structure and energy transport (Nicholls et al. (2024)). Rather than time-stepping each model level according to radiative heating and applying convective adjustment (e.g. Malik et al. (2017), Selsis et al. (2023), Pierrehumbert (2010)), AGNI uses the Newton-Raphson method to conserve energy fluxes through the column to a required tolerence. This is similar to the method applied by Drummond et al. (2016) and Goyal et al. (2020). This means that the model takes tens or hundereds of iterations to obtain a solution, in comparison to thousands or tens of thousands.

Radiative transfer is performed under the correlated-k and two-stream approximations using SOCRATES[3], a well established FORTRAN code developed by the UK Met Office (Amundsen et al., 2014; Amundsen, David S. et al., 2017; Manners & others, 2024; Sergeev et al., 2023). Convection, condensation, and sensible heat transport are also modelled.

AGNI was used in Hammond et al. (2024).

[2]The PROTEUS framework can be found on GitHub here.
[3]SOCRATES is mirrored on GitHub, with additional tools, under its original open-source license here.

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